DATA ITEM DESCRIPTION

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1. TITLE	2. IDENTIFICATION NUMBER
AIRCRAFT MONITOR AND CONTROL (AMAC) DESIGN ANALYSIS REPORT	DI-NUOR-81408

3. DESCRIPTION/PURPOSE

3.1 This report provides a detailed analysis of the Aircraft Monitor and Control (AMAC) system design for nuclear weapons supporting the system interface specification conformance analysis.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
940729	F/SA-ALC-NWI		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the tasks described by 90.2 of MIL-STD-1822.

7.2 This DID is related to DI- NUOR-81409 , Nuclear Certification Plan, which defines the preparation task and submittal schedule for the Design Analysis Report (DAR).

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		F7045

10. PREPARATION INSTRUCTIONS

10.1 Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

10.2 Format. Contractor format is acceptable.

10.3 Content. The report shall contain a complete design description and analysis of the AMAC system as installed on the operational aircraft. The detailed analysis shall include hardware and software designs. The analysis shall demonstrate compliance with the appropriate AMAC system interface specifications criteria. The report shall include the following:

10.3.1 Introduction. The introductory section shall describe the requirement(s) that the DAR will satisfy and the purpose and scope of the DAR (i.e.,any limitations on the description or analysis of the AMAC system). The introduction should briefly describe the contents of each section, addendum and appendix.

10.3.2 Applicable documents. This section shall contain the following information:

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11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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10.3.2.1 Applicability. This subsection defines the relevancy of the documents listed in the applicable documents section and the hierarchy of the applicable documents.

10.3.2.2 Government documents. This subsection defines the applicable government reference documents that are pertinent to the descriptions and analyses in the DAR.

10.3.2.3 Contractor's documentation. This subsection defines the contractor's reference documentation that is applicable to the descriptions and analyses presented in the DAR.

10.3.3 AMAC configuration. This section shall describe the aircraft hardware and processor software incorporated in the AMAC system. This section shall contain the information defined in the following paragraphs:

10.3.3.1 Overview. This subsection gives a functional overview of the complete AMAC system. This subsection briefly describes the role of each component in the AMAC system and the relative dedication of the component's function to the AMAC system function (i.e., a nuclear weapon's interface unit is more dedicated to the AMAC function than is the pilot's weapons display). Special points to consider in this subsection are: whether AMAC control in this aircraft is hardwired, whether the AMAC function is distributed throughout the avionics system or not, and the method of communications between all of the various components of the AMAC system (i.e., the message protocol involved).

10.3.3.2 Aircraft avionics nuclear configuration. This subsection describes the nuclear configuration of the aircraft's avionic system. The nuclear configuration consists of the hardware and processor software that directly or indirectly control and monitor the nuclear weapon interfaces. The components of the avionics system that are defined as the nuclear configuration must be subjected to the controls necessary to maintain nuclear weapon compatibility throughout the service life of the aircraft system. This subsection describes as part of the nuclear configuration any nuclear input devices and displays the aircrew uses to monitor the status of nuclear weapons or control nuclear weapons. This subsection includes a detailed text description of the function of each component of the nuclear configuration with figures and diagrams used to clarify the text portion, but not in lieu of a text description.

10.3.3.3 System hardware configuration. This subsection describes the avionics hardware that constitutes the AMAC system of the aircraft. It describes the avionics hardware in the aircraft's navigation subsystem, weapons delivery subsystem, controls and displays subsystem, and the stores management subsystem as they relate to the AMAC system function. This subsection also describes the aircraft's avionics processors and their role in the AMAC system.

10.3.3.4 System software configuration. This subsection describes the avionics flight software that participates in the tasks of controlling and monitoring the nuclear weapon interfaces. If the avionics flight software is stored on a nonvolatile media and then loaded into the avionics processor for operation, this subsection describes the transfer process and the initialization of the avionics processors until the avionics system is fully functional. This subsection describes the avionics processor software modules that interface directly or indirectly to the AMAC functions. This subsection also describes the avionics system executive software, the weapon delivery system software, and the controls and displays software as they relate to the AMAC system function. Isolation of the AMAC functions will also be described.

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10.3.4 AMAC design analysis. This section shall contain the information described in paragraphs 10.3.4.1 through 10.3.4.4 below. The first and second subsections shall use the system hardware and software configurations to define where the appropriate AMAC requirements are satisfied and where they are not. The third and fourth subsections shall take a systems approach in the analysis of the avionic system to follow all design constraints in monitoring and controlling nuclear weapons. For the purposes of this DID, a systems approach is defined as tracing each signal that goes into or comes out of the nuclear weapons connector through each hardware and software component of the AMAC functions that are integrated into the avionics system.

10.3.4.1 System hardware design analysis. This subsection shall describe the hardware interaction and the sequence of events that take place from the time the aircrew member initiates an operator input to the controls and displays subsystem of the aircraft's avionics, through the hardware interfaces, until that operator's input results in a signal which appears at the nuclear weapon interface. This design analysis shall cover all of the hardware components as they relate to each system interface requirement as a group, if the AMAC function are distributed throughout various avionic subsystems.

10.3.4.2 System software design analysis. This subsection shall provide an analysis of the design of the avionics software as it relates to the AMAC of nuclear weapons. It shall also provide an analysis of the implementation of the authorization, safing, prearming, launch, and release signals in the avionic systems software (this includes avionic software executive program). The analysis shall describe how the implementation meets or fails the design criteria of MIL-STD-1822. The analysis shall provide any underlying rationale or criteria used, if the analysis is used as the basis to recommend certification of the software.

10.3.4.3 System monitor design analysis. This analysis shall describe the nuclear weapon monitor function as it exists in the aircraft's avionics system. The analysis shall show the software and hardware signal path from the aircrew members' input into the AMAC system to the response of that input at the weapon's interface. The analysis shall also show the software signal path in the context of the hardware and physical interconnections that the software signals sequence through and the actions that the system hardware takes in response to those software commands.

10.3.4.4 System control design analysis. This subsection shall describe the nuclear weapon control function as it exists in the aircraft's avionics system. The analysis shall show the software signal path in the context of the hardware and physical interconnections that the software signals sequence through and the actions that the system hardware takes in response to those software commands. This subsection shall show the software and hardware signal path from the aircrew member's input into the AMAC system to the response of that input at the weapon's interface.

10.3.5 Aircraft suspension and release equipment. This section shall describe and analyze the suspension and release equipment used on the aircraft and their relationship to the AMAC function. The requirements of this section pertain to internal aircraft suspension and release systems, like rotary launchers, and to the external carriage of nuclear munitions on pylons. The hardware and software analysis of the suspension and release equipment shall be the same as section 10.3.4.

10.3.6 Test equipment. This section shall describe the test equipment used in the testing of the nuclear configuration components of the avionics system, weapon preload testing, and other nuclear weapon interface testing. This section shall also describe the configuration (hardware and software) of automatic

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testers used in the avionics testing, the type of testing, and the pass or fail criteria of the unit when testing nuclear configured line replaceable units. This section shall also describe the steps taken to prevent degradation of nuclear capability of the avionics when using automatic testers.

10.3.7 Appendices or addenda. To keep the main body of the DAR as clear as possible appendices or addenda shall be used to present supporting design material. Information contained in existing documents shall be referenced rather than repeated. The following supporting design material is required:

a. Avionics system (diagrams and text).

b. Interface definition for the weapons interface unit(s) and the interface specifications for all interfaces that the software signal path analysis defines.

c. Timelines (timing sequences and timing diagrams as they relate to the AMAC functions and release of the nuclear weapons).

d. Maintenance logic diagrams (for the key components of the nuclear weapon delivery system).

e. Nuclear weapon delivery system line replaceable unit schematics.

f. Software compatibility (listing Computer Program Identification Numbers (CPINs) for the avionics flight software versions currently in use at operational bases). The following items should also be included:

1. Software signal path definition diagrams.

2. An analysis of the AMAC compatibility of each software package listed in the CPIN list. This shall include:

(a) Software problems and analysis including technical order work arounds, as appropriate.

(b) A review and analysis of all software patches and changes to the source code since the last recompile.

(c) A summary of software verification requirements, test requirements, test results, and any test limitations.

3. A description of all software documentation pertinent to the current avionics flight software packages in use at operational bases.

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